

CHEMICAL EQUILIBRIUM

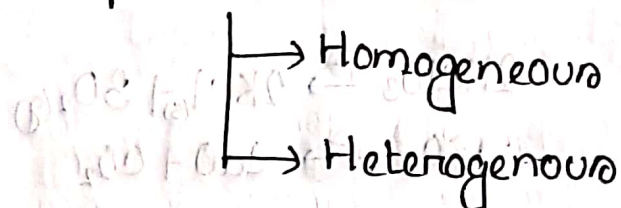
Subject / Theme :

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- Forward Reaction
- Backward Reaction
- Irreversible Reaction
- Reversible Reaction

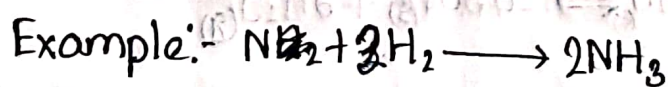
→ Types of chemical equilibrium



- Characterisation of chemical equilibrium
- Law of mass action and equilibrium constant.
- Relation between K_p and K_c
- La Chatelier Principle.

☐ Forward Reaction: →

A forward reaction is in which products are produced from reactants and it goes from left to right in a reversible reaction.

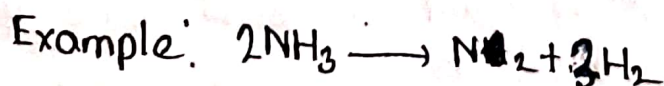


i.e. reactants → products.

☐ Backward Reaction: ←

A backward reaction is a reaction in which reactants are produced from products and it goes from right to left in a reversible reaction.

i.e., Reactants ← Products.

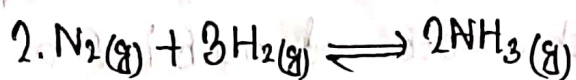
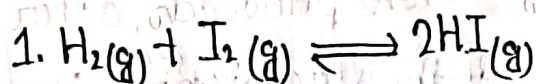


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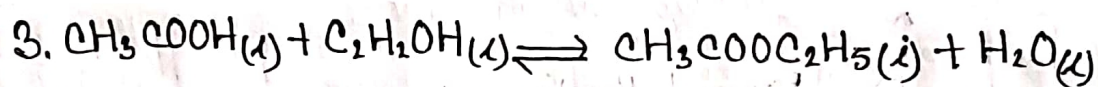
Homogenous Equilibrium:-

The chemical equilibrium in which all reactants and products are in the same physical state is called homogenous equilibrium.

Example:-



All products and reactants are in same gaseous state

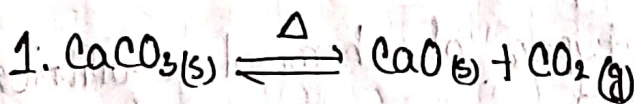


All products and reactants are in liquid state.

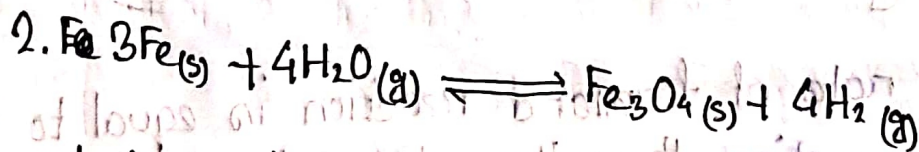
Heterogeneous Equilibrium:-

A chemical equilibrium in which at least one product or reactant is in different physical state (or phase) than the others, is called a heterogeneous equilibrium.

Example:-



The two components are in solid state and one component in gaseous state.



One product and one reactant in solid state but one reactant and one product in gaseous state.

☐ Characteristics of Chemical Equilibrium:-

There are some characteristics of chemical Equilibrium. They are:-

1. **Reversibility of reaction:** Chemical equilibrium is related to only reversible reactions. It is not applicable to irreversible reactions.
2. **Stability of equilibrium:** If the external conditions (Pressure, temperature, and concentration) are not changed, the system will stay there forever.
3. **Easy approachability from both sides:** The equilibrium can be attained from both directions.
4. **Incompleteness of reaction:** A reversible reaction is never complete, because due to backward reaction, the initial reactants are always reformed.
5. **Ineffectiveness of catalysts:** The catalysts have no effect on chemical equilibrium. Catalysts speed up both the forward and backward reaction to the same degree.
6. **Closed System:** It is only attained in a closed system. So that part of any product or reactant cannot escape out.
7. **Dynamic Nature:** Chemical equilibrium is dynamic. Although apparently reaction stops when chemical equilibrium is attained.

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But in reality the forward and backward reactions continue to occur but since their rates are equal & no change is visible.

☐ Law of mass action and equilibrium constant:-

In 1867 mathematician C.M. Guldberg and Chemist P.

Waage of Norway invented a law regarding the rate of chemical reactions, which became known as law of mass action.

The law is as follows:-

At a constant temperature the rate of a chemical reaction at any instant is directly proportional to the active mass (i.e. molar concentration or partial pressure) of the reactants at that instant participating in the reaction.

☐ Le Chatelier Principle:-

If a change occurs in one of the controlling factors, such as temperature, pressure, concentration etc. under which a system is in equilibrium the system will tend to adjust itself in such a way so as to reduce the effect of that change.

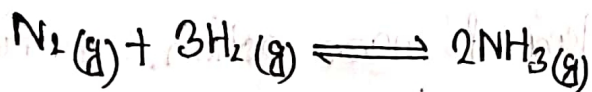
There are three ways in which the stress can be caused on a chemical equilibrium.

1. Changing the concentration of reactant or product.
2. Changing the pressure (or volume) of the system.
3. Changing the temperature.

If a change in concentration, pressure or temperature is caused to a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimize the change.

Effect of change in concentration:-

When concentration of any of the reactants or products is changed, the equilibrium shifts in a direction so as to reduce the change in concentration that was made.



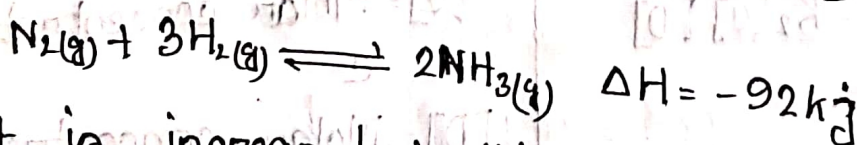
When N_2 or H_2 is added to the equilibrium stage, the equilibrium will shift to the right side so ~~that~~ as to reduce the concentration of the N_2/H_2 .

To have a better yield of NH_3 , one of the reactants should be added in excess.

Effect of change of temperature:-

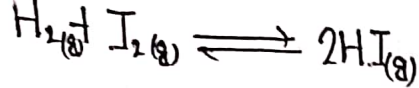
When temperature of a reaction is increased the equilibrium shifts in a direction in which heat is absorbed.

Ammonia synthesis reaction is an exothermic reaction.



As heat is increased in this reaction, an increase of reaction will cause the equilibrium to shift to the left and NH_3 production will be decreased.

Exothermic Reaction



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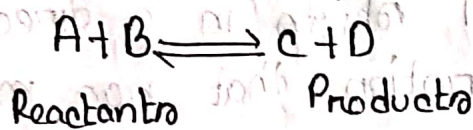
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Effect of change of Pressure:-

When pressure is increased on a gaseous equilibrium reaction, the equilibrium will shift in a direction which tends to decrease the pressure.

Equilibrium Constant:- [साम्य स्थिरांक]

Let us consider following reaction,



According to the law of mass action, the rate of forward reaction,

$$R_f \propto [A][B]$$

$$R_f = K_1[A][B]$$

where,

$[A]$ = Molar concentration of the reactant A

$[B]$ = Molar concentration of the reactant B.

K_1 = Rate constant of the forward reaction

Similarly the rate of Backward reaction,

$$R_b \propto [C][D]$$

$$R_b = K_2[C][D]$$

where,

$[C]$ = Molar concentration of the product C

$[D]$ = Molar concentration of the product D

K_2 = Rate constant of the backward reaction.

At the equilibrium, the rate of forward and backward reaction will be same,

$$R_f = R_b$$

$$\Rightarrow K_1 [A] [B] = K_2 [C] [D]$$

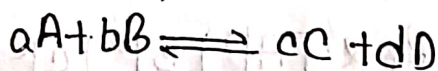
$$\Rightarrow \frac{[A] [B]}{[C] [D]} = \frac{K_2}{K_1}$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[C] [D]}{[A] [B]}$$

$$\Rightarrow K_c = \frac{[C] [D]}{[A] [B]}$$

$\therefore K_c$ is called the molar equilibrium constant or simply equilibrium constant of the reaction,

$K_c = \frac{[C] [D]}{[A] [B]}$ is the mathematical expression for the law of chemical equilibrium.



Forward reaction,

$$R_f \propto [A]^a [B]^b$$

$$R_f = K_1 [A]^a [B]^b$$

Backward reaction,

$$R_b \propto [C]^c [D]^d$$

$$R_b = K_2 [C]^c [D]^d$$

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At equilibrium,

$$R_f = R_b$$

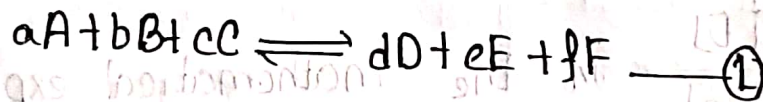
$$\Rightarrow K_1 \cdot \frac{[A]^a [B]^b}{[C]^c [D]^d} = K_2 [C]^c [D]^d$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\Rightarrow K_c = \frac{C_c^c \times C_d^d}{C_A^a \times C_B^b}$$

$$\Rightarrow K_p = \frac{P_c^c \times P_d^d}{P_A^a P_B^b}$$

Relation between K_p and K_c :
Let us consider a general reversible gaseous reaction.



Rate of forward reaction,

$$R_f \propto [A]^a [B]^b [C]^c$$

$$R_f = K_1 [A]^a [B]^b [C]^c \quad \text{--- (2)}$$

Rate of backward reaction,

$$R_b \propto [D]^d [E]^e [F]^f$$

$$R_b = K_2 [D]^d [E]^e [F]^f \quad \text{--- (3)}$$

At equilibrium,

$$R_f = R_b$$

$$\Rightarrow K_1 [A]^a [B]^b [C]^c = K_2 [D]^d [E]^e [F]^f$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c}$$

$$\Rightarrow K_c = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c} \quad \text{--- (4)}$$

$$\Rightarrow K_c = \frac{C_D^d \times C_E^e \times C_F^f}{C_A^a \times C_B^b \times C_C^c} \quad \text{--- (5)}$$

$$\Rightarrow K_p = \frac{P_D^d \times P_E^e \times P_F^f}{P_A^a \times P_B^b \times P_C^c} \quad \text{--- (6)}$$

We know,

$$PV = nRT$$

$$\Rightarrow P = \frac{n}{V} RT \quad \text{--- (7)}$$

$$\Rightarrow P = CRT \quad \text{--- (7)}$$

Putting the value of P in equation --- (6)

$$K_p = \frac{(CRT)_D^d (CRT)_E^e (CRT)_F^f}{(CRT)_A^a (CRT)_B^b (CRT)_C^c}$$

$$= \frac{C_D^d \times (RT)^d \times C_E^e \times (RT)^e \times C_F^f \times (RT)^f}{C_A^a \times (RT)^a \times C_B^b \times (RT)^b \times C_C^c \times (RT)^c}$$

$$= \frac{C_D^d \times C_E^e \times C_F^f (RT)^{d+e+f}}{C_A^a \times C_B^b \times C_C^c \times (RT)^{a+b+c}}$$

$$= \frac{C_D^d \times C_E^e \times C_F^f}{C_A^a \times C_B^b \times C_C^c} \times (RT)^{(d+e+f)-(a+b+c)}$$

$$= K_c RT^{(n_2-n_1)} \quad [\text{from (5)}]$$

$$= K_c RT^{\Delta n}$$

Establish relation between K_p and K_c for the given reaction.

$$\text{(6)} \quad \frac{P_D^d \times P_E^e \times P_F^f}{P_A^a \times P_B^b \times P_C^c} = K_p$$

$$PV = nRT$$

$$P = \frac{n}{V} RT$$

$$\text{(7)} \quad P = CRT$$

Putting the value of P in equation (6)

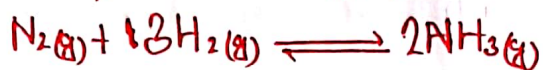
$$K_p = \frac{\left(\frac{n_D}{V}\right)^d \left(\frac{n_E}{V}\right)^e \left(\frac{n_F}{V}\right)^f}{\left(\frac{n_A}{V}\right)^a \left(\frac{n_B}{V}\right)^b \left(\frac{n_C}{V}\right)^c}$$

$$= \frac{\left(\frac{n_D}{V}\right)^d \times \left(\frac{n_E}{V}\right)^e \times \left(\frac{n_F}{V}\right)^f}{\left(\frac{n_A}{V}\right)^a \times \left(\frac{n_B}{V}\right)^b \times \left(\frac{n_C}{V}\right)^c}$$

$$= \frac{C_D^d \times C_E^e \times C_F^f}{C_A^a \times C_B^b \times C_C^c} \times (RT)^{\Delta n}$$

Math:-

1. Some nitrogen and hydrogen gases are pumped into an empty five liter glass bulb at 500°C . When equilibrium is established, 3.00 moles of N_2 , 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of K_c for the reaction.



Here,

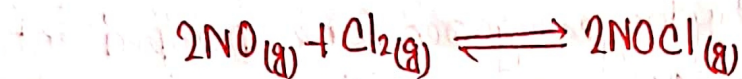
$$[\text{N}_2] = \frac{3.00 \text{ mole}}{5.00 \text{ L}} = 0.6 \text{ M}$$

$$[\text{H}_2] = \frac{2.10 \text{ mole}}{5.00 \text{ L}} = 0.42 \text{ M}$$

$$[\text{NH}_3] = \frac{0.298 \text{ mole}}{5.00} = 0.0596 \text{ M.}$$

$$\begin{aligned} \therefore K_c &= \frac{[\text{NH}_3]^2}{[\text{N}_2] \times [\text{H}_2]^3} \\ &= \frac{(0.0596)^2}{0.6 \times (0.42)^3} \\ &= 0.799. \end{aligned}$$

2. The value of K_p at 25°C for the reaction



is $1.9 \times 10^3 \text{ atm}^{-1}$. Calculate the value of K_c at the same temperature.

Here,

$$K_p = 1.9 \times 10^3$$

$$T = 273 + 25 = 298\text{K}$$

~~Δn~~

$$\Delta n = 2 - (2+1) = -1.$$

$$R = 0.0821$$

We know,

$$K_p = K_c (RT)^{\Delta n}$$

$$\Rightarrow K_c = \frac{K_p}{(RT)^{\Delta n}}$$

$$= \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}}$$

$$= 4.6 \times 10^4$$

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3. The following data was obtained on hydrolysis of methyl acetate at 25°C in 5N hydrochloric acid. Establish it is a first order reaction.

t (secs)	0	4500	7140	∞
ml alkyl used	24.36	29.32	31.72	47.15

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4. In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.70 and 0.91 g respectively. Calculate the transport numbers of the Cu^{2+} and SO_4^{2-} ions.

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Set of drawings